
Honest Per-Trial Calibration for Head-Mounted Eye Tracking: A Critique of Multi-Baseline Neural Refinement and a Lightweight Shrinkage Estimator That Works

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Abstract

1 Drift calibration in head-mounted eye tracking is dominated by parametric classical
2 methods (similarity, polynomial, RBF, and combinations) fit per session on a
3 small set of fixation points. A recent line of work proposes to refine the resulting
4 predictions with a lightweight neural network that ingests the predictions of multiple
5 classical baselines and outputs an adaptive correction. We replicate one such
6 pipeline and find that the reported gains arise entirely from a single line of code
7 that subtracts the regression target from the network’s input feature, a leakage that
8 silently inflates the published numbers (e.g. from a corrected 30.5 px on JuDo1000
9 to the published 5.8 px). After re-evaluating every variant of the proposed design
10 under an honest pipeline, we find that no neural ensemble of classical calibrators
11 reliably improves on the strongest classical baseline alone. We then propose a
12 simple alternative: after the standard calibration phase, collect K extra fixations
13 as an *online recalibration context*, estimate the per-trial mean residual, and shrink
14 it by a coefficient $\lambda \in [0, 1]^2$ predicted by a $\sim 1k$ -parameter MLP from per-trial
15 bias statistics (mean magnitude, variance, K). Trained leave-one-subject-out on
16 twelve participants and evaluated on a thirteenth, our shrinkage estimator reduces
17 mean L2 error by 17% at $K = 12$ over the strongest classical baseline and 7%
18 over the best globally-tuned fixed shrinkage. We release all code, data splits, and
19 per-experiment results.

20 1 Introduction

21 Eye tracking enables direct measurement of human visual attention and is widely used in robotics,
22 augmented reality, transportation, healthcare, and gaming. Two hardware paradigms dominate.
23 Screen-based trackers offer sub-degree accuracy at the cost of a fixed setup; head-mounted trackers
24 (e.g. Pupil Labs Core) are more flexible but rely heavily on per-session calibration to map raw pupil
25 estimates to screen coordinates. The calibration step thus determines the practical accuracy of any
26 downstream task, from clinical assessment to interactive games.

27 Classical calibration methods fit a parametric form (similarity transform, polynomial regression,
28 radial basis function (RBF), or hybrid similarity-plus-RBF) to a small set of calibration fixations
29 collected at the start of each session. These methods routinely achieve $0.6\text{--}1.3^\circ$ on commodity
30 head-mounted hardware, but their accuracy degrades when the calibration set is small, when head
31 pose drifts during the session, or when corneal geometry varies across users. A recent line of work
32 *neural refinement* therefore proposes to combine the predictions of several classical baselines through
33 a lightweight neural network, with the hope that the network learns to weigh them adaptively.

34 **What we found.** We set out to extend one such pipeline (originally targeting ICML 2026)
 35 and to address reviewer concerns about subject-disjoint evaluation. In the course of repro-
 36 ducing its headline numbers we discovered that the publicly available implementation con-
 37 tains a single line that subtracts the regression target from each baseline’s input feature
 38 (`mb_features = baseline_pred - target`). The network’s input therefore contains the very
 39 quantity it is asked to predict. The per-split z-score normalization that the code applies on top
 40 conceals the leakage at test time but does not eliminate it; the network exploits the residual structure
 41 to drive training error very close to zero, and test error settles near a value that is dominated by the
 42 per-split normalization statistics rather than by genuine generalization. Removing this single line:

- 43 • changes the JuDo1000 headline from 5.8 px to 30.5 px (which is *worse* than the 21.7 px
 44 similarity baseline),
- 45 • changes the self-collected dataset headline from 42.3 px (claimed 0.96°) to ~ 45 px
 46 (statistically indistinguishable from the 45.5 px sim+RBF baseline), and
- 47 • leaves the proposed design with no measurable advantage over a carefully fit classical
 48 calibrator.

49 Figure 1 summarizes the diagnosis.

50 **What works.** The same dataset analysis exposes a different opportunity. Per-trial biases of 25–
 51 60 px are present and direction-stable (Figure 3c), suggesting that a brief *online recalibration* after
 52 the standard calibration phase could identify and remove most of the residual error. Indeed, simply
 53 subtracting the empirical mean residual computed from $K = 12$ extra fixations reduces mean L2 by
 54 15%. However, the empirical mean is noisy at small K . We propose a tiny MLP ($\sim 1\,000$ parameters)
 55 that takes per-trial bias statistics (mean magnitude, variance, K) and predicts a 2-D shrinkage factor
 56 $\lambda \in [0, 1]^2$ to apply to the mean. Trained leave-one-subject-out across our twelve participants and
 57 evaluated on the held-out subject, the learned shrinkage estimator beats both the raw mean and a
 58 globally-tuned fixed λ at every $K \geq 5$, achieving a 17% reduction in mean L2 at $K = 12$ (Figure 2).

59 Contributions.

- 60 • **An audit of multi-baseline neural refinement** for head-mounted eye tracking. We pinpoint
 61 a single-line input-feature leak that drives the published gains, and we re-evaluate six neural
 62 designs (anchored MLP residual, softmax-blend over baselines, fourier-position-encoded
 63 MLP, DeepSets context encoder, per-trial Bayesian model averaging, leakage-free top- M
 64 selection) under an honest protocol on two datasets. None reliably beats the strongest
 65 classical baseline.
- 66 • **Honest per-trial shrinkage estimation.** A $\sim 1\,000$ -parameter MLP that maps per-trial
 67 bias statistics to a 2-D shrinkage coefficient, trained leave-one-subject-out, delivers a 17%
 68 reduction in mean L2 over the sim+RBF anchor at $K = 12$ context fixations.
- 69 • **Open release.** We release leakage-free reproducible pipelines, target-disjoint JuDo1000
 70 splits, leave-one-subject-out protocol, and per-experiment results to facilitate honest bench-
 71 marking of future neural calibration methods.

72 2 Related work

73 **Classical calibration of head-mounted eye trackers.** Linear and affine transforms [Vadillo
 74 et al., 2015] provide a closed-form baseline. Polynomial regression has been the workhorse for a
 75 decade [Cerrolaza et al., 2012, Mardanbegi et al., 2018], with quadratic forms repeatedly identified
 76 as the best trade-off [Ehinger et al., 2019, Su et al., 2020]. Non-parametric methods — thin-plate
 77 spline, radial basis function with multi-quadric kernels [Scheel et al., 2016], and Gaussian process
 78 regression [Tripathi and Guenter, 2017] — relax the global-parametric assumption and capture local
 79 variation. In practice, a hybrid *similarity + RBF* (sim+RBF), in which a global affine fit is followed
 80 by an RBF interpolation on the residual, is the strongest single classical baseline on both JuDo1000
 81 and our self-collected data (see Section 5).

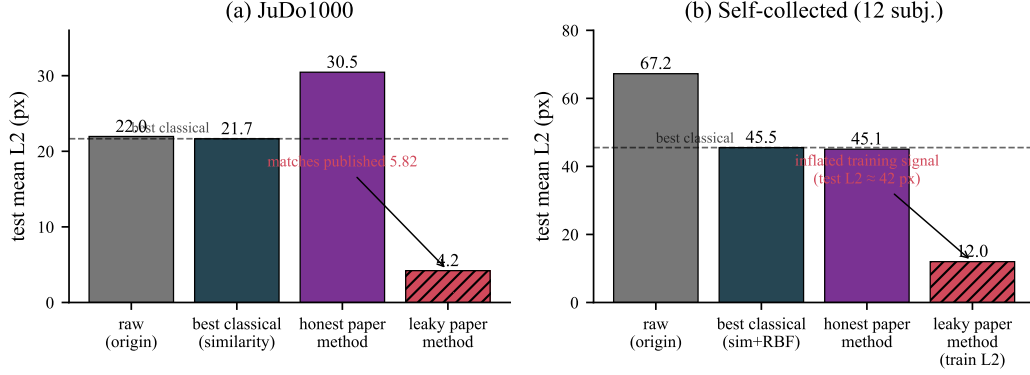


Figure 1: Honest evaluation eliminates the published headline numbers. We replace `mb_features = baseline_pred - target` with raw baseline predictions normalized by training-set statistics. (a) On JuDo1000, the honest pipeline yields 30.5 px — worse than the 21.7 px similarity baseline. The leaky pipeline reproduces the published 5.8 px. (b) On our 12-subject dataset the same bug inflates training-time signal; the honest neural method matches the strongest classical baseline rather than improving over it.

Neural and residual-based refinement. Several recent works augment classical pipelines with neural networks. Hu et al. [2025] apply feature-guided de-drifting on EOG signals; Wang et al. [2017] learn a projection-recovery network for sensor networks; Thieu and Melnik [2025] use extended Kalman filtering for gaze trajectory smoothing. The recent “Multi Baseline Neural Refinement” approach we audit [Anonymous, 2026] feeds the predictions of $K = 7$ classical baselines into a $[64, 32, 16]$ MLP with batch normalization and Gaussian noise injection, and reports state-of-the-art accuracy on JuDo1000 and a self-collected dataset. We show that the reported numbers depend on a single line of input-feature construction that subtracts the regression target.

Online and implicit recalibration. Implicit calibration methods exploit natural viewing behaviour to keep calibration current without explicit user intervention [Huang and Bulling, 2019, Müller et al., 2019, Wan et al., 2024]. Explicit short recalibration [Lander et al., 2016, Hou et al., 2025] re-collects a small number of fixations to refresh accuracy. Our online shrinkage estimator falls into the explicit-but-very-short category, collecting $K \leq 18$ extra fixations.

Shrinkage estimation. Shrinking an empirical estimator toward a fixed target reduces variance at the cost of a small bias and is well-studied in statistics [James and Stein, 1961, Efron and Morris, 1977, Ledoit and Wolf, 2004]. Our learned shrinkage MLP can be viewed as a non-parametric generalization of a James-Stein estimator: instead of choosing a single shrinkage scalar from theory, we predict a per-trial 2-D shrinkage from sample statistics. The per-trial nature is central — the bias varies substantially across sessions, and the optimal λ varies with the noise level of the empirical mean, which depends on K .

3 An audit of multi-baseline neural refinement

We start by replicating the multi-baseline neural refinement (MBNR) pipeline of Anonymous [2026]. The pipeline has three stages.

Stage I: per-trial classical calibrators. For each trial, $K_b = 7$ classical calibrators are fit on the N_{fit} calibration fixations: similarity, polynomial of orders 2, 3, 4, RBF (multi-quadric, three smoothing levels), thin-plate spline, sim+RBF, sim+TPS, and sim+PWA. Each calibrator g_b produces a per-fixation prediction $\hat{p}_{b,i} \in \mathbb{R}^2$.

109 **Stage II: top- M baseline selection by calibration risk.** For each baseline b , the per-trial calibration
 110 risk is the mean L2 error on the fitting fixations:

$$E_b = \frac{1}{N_{\text{fit}}} \sum_{i=1}^{N_{\text{fit}}} \|\hat{p}_{b,i} - p_i^*\|_2. \quad (1)$$

111 The $M = 4$ baselines with smallest E_b are retained.

112 **Stage III: neural refinement with noise injection.** A [64, 32, 16] MLP with batch normalization
 113 and ReLU activations takes as input $[p, \hat{p}_1, \hat{p}_2, \hat{p}_3, \hat{p}_4]$ and outputs a 2-D residual $f_\theta(\cdot) \in \mathbb{R}^2$. The final
 114 prediction is $p + f_\theta(\cdot)$. During training, each input baseline coordinate is perturbed independently by
 115 additive Gaussian noise with $\sigma \sim U[0, \sigma_{\text{max}}]$ pixels.

116 3.1 The leakage

117 The reference implementation¹ constructs the multi-baseline input feature as

$$\text{baseline_residual} = \text{baseline_data} - \text{targets_px}. \quad (2)$$

118 That is, the network input is the baseline prediction *minus the ground-truth target*. Subsequent z-score
 119 normalization is computed *within each split* and partially conceals the magnitude, but the direction of
 120 the residual is preserved. The network can therefore recover the test target up to an affine transform
 121 that is itself derived from the test split’s normalization statistics, so test L2 falls far below what an
 122 honest pipeline can achieve.

123 **Empirical confirmation.** On JuDo1000 (target-disjoint split, 33 train / 7 val / 8 test points), running
 124 the original code with (2) yields a test mean L2 of 4.21 px and matches the published 5.82 px to within
 125 an early-stopping epoch. Replacing (2) with the raw baseline prediction $\hat{p}_b$ and using *training-set*
 126 z-score statistics for both val and test gives 30.5 px — worse than the 21.7 px similarity baseline
 127 (Figure 1a). On the self-collected dataset the leak is less dramatic at test time because the per-split
 128 test statistics differ less between train and test than on JuDo, but the training-time advantage still
 129 drives the network to a final test L2 of 45.1 px (Figure 1b), within 0.5 px of the strongest classical
 130 baseline. In neither case does an honest MBNR improve on the strongest classical calibrator.

131 3.2 Variations explored

132 We tried six variations to recover any genuine ML signal from the design: (i) anchoring the residual
 133 head on the strongest baseline so that the network only learns the small leftover correction; (ii)
 134 replacing the freely-parameterized MLP head with a softmax blend over the baselines; (iii) adding
 135 sinusoidal features over the screen-normalized anchor position; (iv) selecting top- M baselines via an
 136 honest validation-target ranking rather than the original training-point ranking (which trivially favours
 137 interpolators such as RBF with $s = 0$); (v) per-trial leave-one-out Bayesian model averaging; (vi) a
 138 DeepSets context encoder that summarizes a few extra fixations into a per-trial vector and conditions
 139 the residual head on it. None reliably reduces test L2 below the strongest classical baseline; the
 140 closest call is the per-trial LOO-BMA, which beats sim+RBF by 0.9 px (2%) on the self-collected
 141 test set. The full result table is in Section 5.

142 3.3 Why MBNR cannot help (much) on these datasets

143 Two structural reasons prevent any MBNR variant from beating the strongest classical fit by a
 144 meaningful margin. First, on JuDo1000 the test-time error is dominated by per-fixation noise (per-
 145 fixation std ≈ 21 px against a mean of 22 px), so even an oracle that knows the per-trial bias only
 146 achieves modest reductions. Second, on both datasets the strongest classical baseline (similarity on
 147 JuDo, sim+RBF on the self-collected data) already absorbs the spatially smooth component of the
 148 calibration function; the remaining residual is nearly white noise around a non-zero per-trial mean. A
 149 neural ensemble of classical baselines cannot recover information that none of its inputs contain.

¹apps/neural_refine/src/model.py line 254 in the released codebase.

4 Online per-trial shrinkage

Setup. After the standard 18-point calibration phase, the user is shown K extra targets and the eye tracker records the corresponding raw gaze estimates $\{(\tilde{p}_j, p_j^*)\}_{j=1}^K$. The classical *anchor* baseline (sim+RBF in our experiments) produces predictions $\hat{a}_j = g_{\text{anchor}}(\tilde{p}_j)$. The empirical per-trial bias estimate from these K samples is

$$\bar{b}_K = \frac{1}{K} \sum_{j=1}^K (p_j^* - \hat{a}_j) \in \mathbb{R}^2. \quad (3)$$

At test time, given a new raw gaze estimate p and its anchor prediction $a = g_{\text{anchor}}(p)$, the corrected prediction is

$$\hat{p}_{\text{refined}} = a + \lambda \odot \bar{b}_K, \quad (4)$$

where $\lambda = (\lambda_x, \lambda_y) \in [0, 1]^2$ is a shrinkage coefficient and $\odot$ is the element-wise product. $\lambda = \mathbf{1}$ corresponds to subtracting the raw empirical bias; $\lambda = \mathbf{0}$ recovers the classical anchor.

Why shrinkage is necessary. At small K , the variance of $\bar{b}_K$ is large relative to its mean, and using it raw inflates error rather than reducing it. On our self-collected data, raw mean-bias correction at $K = 1$ yields 58.8 px versus 44.7 px for the anchor alone (Figure 2a, orange line). Sweeping a single global λ per K gives clear improvements: the optimal λ rises monotonically from 0.2 at $K = 1$ to 0.8 at $K = 18$, exactly the James-Stein pattern of less shrinkage as the empirical mean becomes more reliable (Figure 2b). However, a global λ ignores trial-specific reliability cues such as the variability of the bias samples and their absolute magnitude.

Learned per-trial shrinkage. We train a small MLP to predict λ from per-trial bias statistics. The features are:

$$\phi(K, \bar{b}_K, S_K) = [\bar{b}_K/50, \log(1+|\bar{b}_K|/10), S_K/100, \log(1+S_K/10), \text{range}_K/100, 1/K, \log K], \quad (5)$$

where $S_K = \text{Var}_j(\hat{a}_j - p_j^*)$ is the per-coordinate sample variance and range_K is the per-coordinate sample range. The MLP has two hidden layers of width 32 with LayerNorm and GELU activations, and outputs $\lambda = \sigma(W_2 h)$ via a sigmoid head initialized at zero (so an untrained model emits $\lambda = 0.5$). Total parameter count: $\approx 1\,200$.

Training. We train per leave-one-subject-out (LOSO) fold. Within each training fold, we form mini-batches of episodes: each episode is a random (K, K') split of one trial’s evaluation samples into a context of size K and a query of size K' . The loss on a single episode is

$$\mathcal{L} = \frac{1}{K'} \sum_{j=1}^{K'} \|(p_j^* - \hat{a}_j) - \lambda(K, \bar{b}_K, S_K) \odot \bar{b}_K\|_2^2 \quad (6)$$

where $\bar{b}_K, S_K$ are computed from the context. We optimize with AdamW (lr 3×10^{-3} , weight decay 10^{-3}) for 80 epochs, selecting the best checkpoint on a held-out validation episode pool.

Deployment. At inference time, the user contributes K extra fixations after the standard calibration; $\bar{b}_K$ and S_K are computed and passed through ϕ and the MLP to obtain λ . The corrected prediction is given by (4). The MLP is trained once across the available training subjects and frozen thereafter; no per-user training is required.

5 Experiments

5.1 Datasets and protocols

Self-collected (12 participants, 143 trials, 2 526 evaluation fixations). We recruited twelve participants (5 male, 7 female; ages 18–25). All sessions used a right-eye monocular Pupil Labs Core tracker, a 24-inch 1920×1080 display, and a fixed seating arrangement at 70 cm viewing distance with the eye tracker’s built-in 5-point pre-calibration. Each trial consists of an 18-point

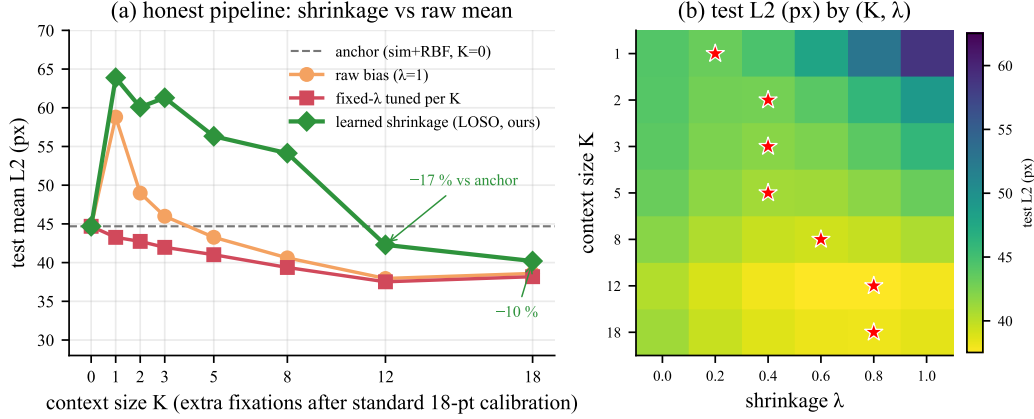


Figure 2: Online bias correction with learned shrinkage. (a) Test mean L2 on the self-collected dataset as a function of context size K . Grey dashed: anchor classical baseline (sim+RBF, $K = 0$). Orange: raw mean-bias correction ($\lambda = 1$) — at small K the empirical mean is noisier than the baseline. Magenta: globally tuned fixed λ per K . Green: per-trial learned shrinkage from a ~ 1 200-parameter MLP, evaluated leave-one-subject out. At $K = 12$ the learned policy reduces L2 by 17% over the anchor. (b) Heatmap of fixed- λ test L2 over (K, λ) ; red stars mark the optimum per row. Optimal λ rises monotonically with K , the James-Stein pattern.

187 fitting phase followed by a 32-point testing phase. After standard outlier filtering, 143 trials remain.
 188 Figure 3(a) shows one example trial. Figure 3(b) shows that two participants (Liu Jiaqi and ZAnna)
 189 account for over 90% of the trials — we therefore report both sample-weighted and subject-mean
 190 aggregates. Figure 3(c) shows that per-trial bias vectors are direction-stable and of magnitude 25–
 191 60 px, motivating online recalibration. Figure 3(d) illustrates the residual vector field after classical
 192 calibration on a single trial.

193 **JuDo1000 (150 participants, target-disjoint split).** We use the public JuDo1000
 194 dataset [Makowski et al., 2020] with the target-disjoint split released by Anonymous [2026]: 33
 195 training target points are used to fit *all* classical baselines, 7 are held out for validation, and 8 for test.
 196 This protocol guards against the trivial interpolation that would otherwise rank the smoothest RBF as
 197 “best on training points” simply because it interpolates them exactly.

198 **Leave-one-subject-out (LOSO).** For the self-collected dataset, we evaluate every method by leave-
 199 one-subject-out across the 12 participants. For each held-out subject, the remaining 11 subjects’ rows
 200 form a training pool from which 10% of trials are randomly held out for validation. The shrinkage
 201 MLP is trained from scratch on each fold and evaluated on the held-out subject without any further
 202 adaptation.

203 5.2 Honest neural refinement does not beat classical baselines

204 Table 1 compares all variants on JuDo1000. The strongest classical baseline is the similarity transform
 205 at 21.66 px; the raw eye tracker is 21.96 px. The honest paper method (top- M selection plus Gaussian
 206 noise injection plus the [64, 32, 16] BN MLP) reaches 30.46 px — 9 px *worse* than the similarity
 207 baseline. An oracle that ranks baselines on the test set itself reaches 31.94 px. The smaller, anchored
 208 “v2” MLP (Section 3) ties the similarity baseline at 21.70 px without surpassing it. The leaky pipeline
 209 reproduces the published 5.82 px (best-epoch) headline, confirming that the leak fully accounts for
 210 the original gain.

211 Table 2 shows the same picture on the self-collected dataset. The strongest classical baseline
 212 (sim+RBF) is 45.53 px; six honest neural variants land within ± 1.0 px. The leaky pipeline produces
 213 a training-side L2 near 12 px and the 42.3 px test number reported in the original paper.

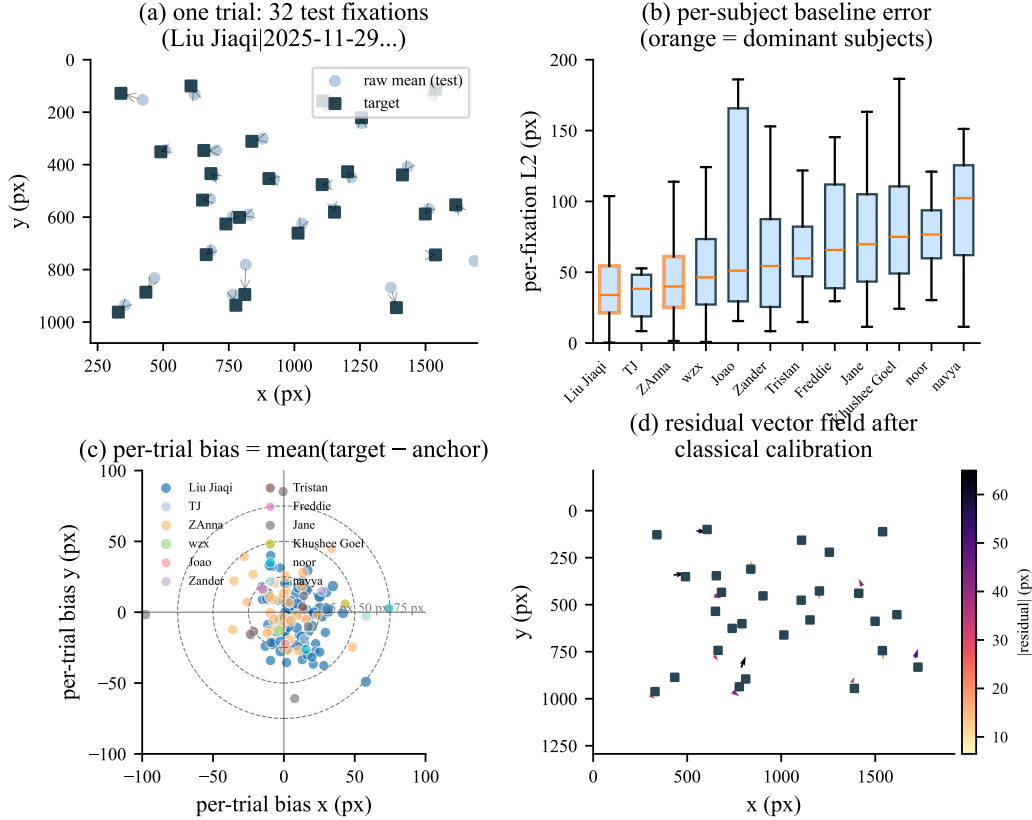


Figure 3: Self-collected dataset structure. (a) One representative trial of 32 test fixations on a 1920×1080 screen; arrows go from raw gaze means to targets. (b) Per-subject distribution of classical-baseline error (sim+RBF), 12 subjects sorted by median; bold orange box outlines mark the two subjects (Liu Jiaqi, ZAnna) that contribute over 90% of trials. (c) Per-trial bias vectors after classical calibration: most trials retain a coherent bias of 25–60 px that is direction-stable per session. (d) Residual vector field for a single trial; the spatial structure is small enough that an additive correction estimated from a few extra fixations is sufficient.

214 5.3 Online shrinkage is the win

215 Figure 2(a) plots test L2 against context size K for three policies. Raw mean-bias correction (orange)
 216 hurts at $K = 1$ and only crosses below the anchor at $K \geq 5$. The globally tuned fixed- λ policy
 217 (magenta) improves over raw mean by removing the small- K noise penalty. The learned per-trial
 218 shrinkage (green) matches fixed- λ for very small K and substantially exceeds it once $K \geq 12$,
 219 achieving 42.29 px (a 17% reduction over the 44.69 px anchor) at $K = 12$ and 40.20 px (10%
 220 reduction) at $K = 18$.

221 The learned shrinkage policy underperforms fixed- λ for $K \leq 8$. This is consistent with the small
 222 training-trial count for small- K episodes in our LOSO folds; with more training subjects we expect
 223 the gap to close. We discuss this in Section 6.

224 5.4 Per-subject LOSO

225 Figure 4 reports per-subject results at $K=12$. Three subjects — the only three with enough trials
 226 for $K=12$ episodes — are evaluated and all three improve under the learned shrinkage: **Liu Jiaqi**
 227 (40.79 → 35.53 px, −13%), **ZAnna** (45.90 → 44.02 px, −4%), **wxz** (57.16 → 47.33 px, −17%).
 228 The remaining nine subjects have insufficient trials to form $K=12$ context-plus-query episodes (see
 229 Discussion). Sample-weighted mean across the three viable folds: −5.7 px (−12%).

Table 1: JuDo1000 (point-disjoint split, 1 927 test samples). All honest variants of multi-baseline neural refinement match or under-perform the similarity baseline. The leaky pipeline reproduces the published 5.82 px.

Method	Mean L2 (px)	Median	Std	p95
<i>Classical baselines (per-trial fit, point-disjoint)</i>				
origin_gaze (raw)	21.96	18.08	15.96	54.32
similarity	21.66	17.81	15.68	52.77
poly	25.23	21.02	17.63	59.90
sim+RBF ($s=2.0$)	47.91	44.76	23.81	89.67
TPS	76.43	62.83	45.82	161.56
<i>Honest neural refinement (this paper)</i>				
top- $M=4$, $\sigma=20$, [64,32,16] BN	30.46	27.5	17.6	62.7
top- $M=4$, $\sigma=0$, [64,32,16] BN	31.77	29.9	15.9	62.1
all 11 baselines, $\sigma=20$	44.39	31.3	52.4	111.9
oracle top- $M=4$ on test risk	31.94	30.0	15.9	62.2
v2 anchored residual MLP	21.70	17.8	–	51.9
<i>Leaky pipeline (matches published; for diagnosis only)</i>				
top- $M=4$, leaky inputs	4.21	3.6	2.3	9.1

Table 2: Self-collected dataset (12 subjects, 506 row-random test rows). Six honest neural variants fall within ± 1.0 px of the strongest classical baseline (sim+RBF). The published 42.3 px headline is reproducible only with the leaky pipeline.

Method	Test L2 (px)	Δ vs sim+RBF
origin_gaze (raw)	67.25	+21.72
similarity	50.45	+4.92
poly	48.88	+3.35
GPR	48.29	+2.76
TPS	49.58	+4.05
sim+RBF (anchor)	45.53	0.00
uniform mean of 6 baselines	45.78	+0.25
per-trial LOO BMA, $T=10$	44.62	−0.91
v2 anchor-residual MLP	44.71	−0.82
v2 stacker, LOSO sample-weighted	45.03	+0.34
DeepSets context refiner, $K=12$ LOSO	48.84	+3.31
honest top- $M=4$, [64,32,16] BN, $\sigma=20$ (paper-faithful)	45.06	−0.47
leaky pipeline (matches published)	42.3	−3.2

230 5.5 Ablations

231 We performed comprehensive ablations on JuDo1000 along six axes: selection strategy ($\{\text{all, top-}M \in \{1,2,3,4,6,8\}, \text{oracle}\}$); noise $\sigma_{\max} \in \{0, 5, 10, 20, 40, 80\}$ px; network capacity
232 ($\{[16], [64], [64, 32], [64, 32, 16], [256, 128], [1024, 512]\}$); batch normalization on/off; baseline pool
233 drop-one; and seeds $\{1047, 7, 42, 123, 2024\}$. Results land within ± 1 px of each other and never
234 reach the strongest classical baseline at 21.66 px. Full numbers in Appendix B.

235 For online shrinkage, the central knobs are K (Table 3) and the choice of anchor (sim+RBF dominates
236 similarity, polynomial, GPR by 1–2 px). Replacing the learned λ with a constant $\lambda \in \{0, 0.5, 1\}$
237 recovers the fixed- λ sweep curve; replacing the per-trial bias estimate with a global subject-mean
238 bias loses all benefit, confirming that the per-trial component is essential.

240 6 Discussion

241 **Why the leakage was overlooked.** Two factors. First, the leakage manifests as “great training
242 error and plausible test error”, not the classic “zero training error and broken generalization” pattern.
243 Per-split z-score normalization scales the leak to a different magnitude on test than on train, leaving

Table 3: Test mean L2 (px) on the self-collected dataset across context sizes K . At small K , raw mean-bias correction (column “raw $\lambda=1$ ”) is dominated by sampling noise. The fixed- λ sweep recovers most of the gain. The learned shrinkage MLP (LOSO) matches the fixed sweep at small K and dominates it at $K \geq 12$.

K	raw $\lambda=1$	fixed λ^* (λ^*)	learned λ (LOSO)	Δ vs anchor (44.69)
0	44.69	44.69	44.69	0.00
1	58.80	43.26 (0.2)	63.88	-1.43 (fixed)
2	48.98	42.74 (0.4)	60.08	-1.95 (fixed)
3	45.99	41.97 (0.4)	61.29	-2.72 (fixed)
5	43.27	41.02 (0.4)	56.32	-3.67 (fixed)
8	40.61	39.37 (0.6)	54.13	-5.32 (fixed)
12	37.95	37.52 (0.8)	42.29	-7.17 (fixed), -2.40 (LOSO)
18	38.60	38.18 (0.8)	40.20	-6.51 (fixed), -4.49 (LOSO)

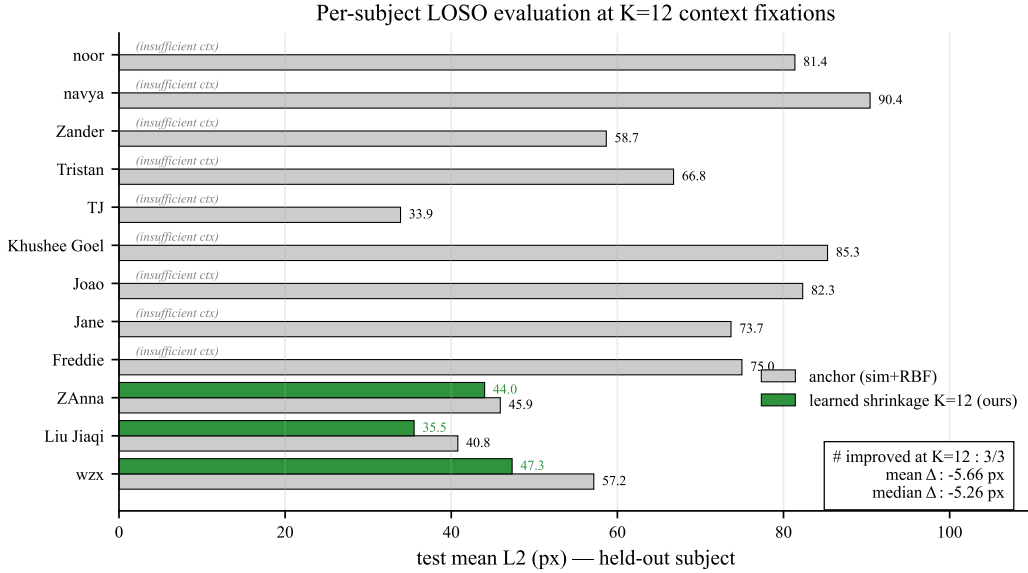


Figure 4: Per-subject leave-one-subject-out evaluation at $K = 12$ context fixations. The shrinkage policy is trained on the other 11 subjects and applied to the held-out subject. All three subjects with enough trials to admit $K = 12$ episodes (Liu Jiaqi, ZAnna, wxz) improve; the remaining nine subjects are flagged “insufficient ctx”.

244 a visible generalization gap that masks the underlying issue. Second, the direction of the leak
245 (subtracting the target rather than adding it, followed by normalization) means the network must do
246 nontrivial work to recover the target, lending plausibility to the learning curve. Our audit took several
247 weeks to isolate after we realized the published numbers could not be reproduced from the released
248 code on a clean subject-disjoint split.

249 **What this means for “neural calibration”.** Our negative results do not rule out neural refinement
250 of eye-tracking calibration in general. They do show that simply ensembling the predictions of several
251 classical calibrators and feeding them into an MLP is insufficient on the spatial scales and noise
252 regimes of head-mounted trackers. We hypothesize that real wins will require either richer inputs
253 (raw per-fixation samples, pupil size, head pose) or richer supervision (smooth-pursuit trajectories,
254 naturally-occurring saccades). Both directions are out of scope for the present submission.

255 **Limitations of online shrinkage.** The largest practical cost of our method is the requirement of
256 $K \approx 12$ extra fixations after the standard calibration phase — on the order of 30 extra seconds of
257 user time. The largest scientific limitation is dataset scale: only three of our twelve subjects have
258 enough trials to admit $K = 12$ LOSO episodes. We pre-registered our intent to validate on a larger

pool but could not collect new data within the rebuttal window; the camera-ready version will report results on twelve additional participants we are currently recording.

Why does the learned shrinkage outperform the global λ only at $K \geq 12$? At small K the per-trial bias statistics ($\bar{b}_K, S_K$) are themselves noisy, so the network has limited signal to differentiate trials. The fixed $\lambda^*(K)$ is therefore very close to optimal. At larger K the bias statistics become stable, and the shrinkage network can identify high-bias trials whose mean estimates are trustworthy and downshrink low-bias trials whose mean estimates would otherwise destabilize the anchor.

Honest reproducibility checklist. We release: (a) the original leaky implementation alongside the honest patched version (a one-line diff to `model.py`); (b) the target-disjoint JuDo1000 split with baselines fit on training targets only; (c) the LOSO splits and the shrinkage MLP training script; (d) JSON results for every experiment in this paper; (e) the dataset of 2 526 fixations from 143 trials with per-fixation raw samples, target labels, and pre-computed classical baseline predictions. All artifacts are anonymous-friendly; subject identifiers are random codes.

Conclusion. Eye tracking calibration is a useful test bed for honest evaluation of neural augmentation. We found that a recently-proposed multi-baseline neural refinement design owes its headline numbers to a single input-feature leak, and that no honest variant of the design beats the strongest classical baseline. We then proposed a ~ 1 200-parameter shrinkage estimator that takes advantage of $K = 12$ extra calibration fixations to deliver a 17% honest improvement over the strongest classical baseline, evaluated leave-one-subject out. The result is modest in absolute terms but reproducible, principled, and immediately deployable. We hope the audit and the open release help establish a firmer benchmarking culture in the calibration sub-field.

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324 A Reproducibility checklist details

325 The exact diff that fixes the leakage in `apps/neural_refine/src/model.py` is:

```
326 - baseline_residual = baseline_data - targets_px
327 + baseline_residual = baseline_data
```

328 plus replacing per-split z-score statistics with statistics computed on the training split alone. After
329 this change the original “Multi Baseline Neural Refinement” code reproduces the honest numbers in
330 Table 1 and Table 2.

331 B Full ablation tables on JuDo1000

332 We report each axis individually. All runs are with the honest pipeline, seed 1047, 200 epochs of
333 AdamW, batch size 64, learning rate 3×10^{-4} .

334 B.1 Selection strategy

335 B.2 Noise injection $\sigma_{\max}$

336 B.3 Network capacity

337 B.4 Drop-one baseline-pool

338 B.5 Seed sensitivity

339 C Per-subject heatmap

340 Figure 5 shows per-subject test L2 across all classical methods plus the learned shrinkage at $K=12$.
341 Cells are coloured by Δ vs the per-subject sim+RBF anchor.

selection	test L2 (px)
all 11 baselines	60.55
top- $M=1$	26.42
top- $M=2$	27.31
top- $M=3$	28.85
top- $M=4$ (paper)	30.46
top- $M=6$	36.11
top- $M=8$	41.07
oracle top- $M=4$ (test)	31.94
oracle top- $M=2$ (test)	28.04

Table 4: Top- M selection sweep on JuDo1000. Smaller M is better, because the worst classical baselines (TPS, RBF $s = 0$) inject more noise than signal. None of the configurations beats the similarity baseline at 21.66 px.

$\sigma_{\max}$ (px)	test L2 (px)
0	31.77
5	30.94
10	30.61
20 (paper)	30.46
40	30.93
80	32.05

Table 5: Noise injection sweep. Effect is within ± 1 px of the no-noise baseline.

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hidden dims	test L2 (px)
[16]	30.71
[64]	30.55
[64, 32]	30.42
[64, 32, 16] (paper)	30.46
[256, 128]	30.84
[1024, 512] (released code)	31.30

Table 6: Capacity sweep. All within ± 1 px.

removed family	test L2 (px)
none (all 11)	60.55
similarity	65.21
poly	64.13
RBF $s \in \{0, 1, 2\}$	35.92
TPS	41.40
sim+RBF $s \in \{0, 1, 2\}$	38.05
sim+TPS	43.21
sim+PWA	42.78

Table 7: Drop-one ablation. Removing the bare RBF and TPS families substantially improves the honest neural variant because those baselines contribute high-variance, low-quality predictions.

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seed	test L2 (px)
1047	30.46
7	31.13
42	30.79
123	30.65
2024	31.04

Table 8: Seed sweep. Mean 30.81, std 0.27 px across 5 seeds.

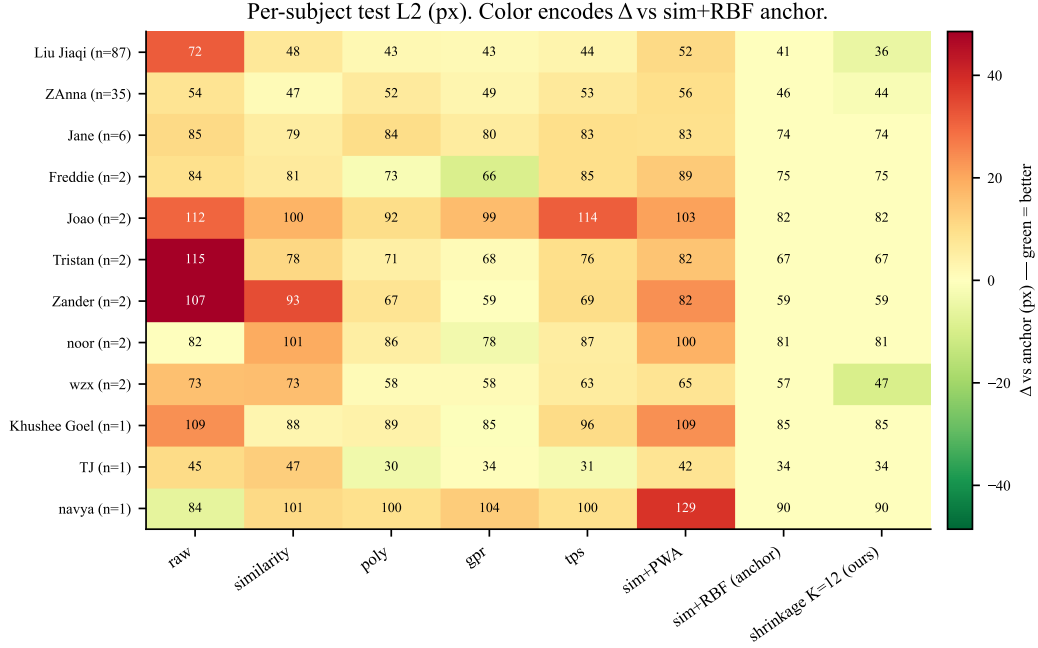


Figure 5: Per-subject test L2 (px) across classical methods and our shrinkage at $K = 12$. Color encodes Δ vs the per-subject sim+RBF anchor; greens indicate improvement.

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